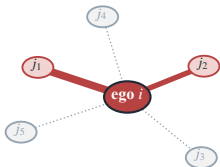


a Social-risk graph • top- $K_s = 2$ by predicted TTC



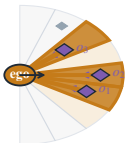
edge thickness $\propto$ attention α_{ij}

risk score r_{soc}

j_1	TTC=1.2s
j_2	TTC=1.6s
j_3	TTC=3.8s
j_4	TTC=4.2s
j_5	TTC=5.0s

selected (top- K_s) • pruned

b Dynamic-obstacle graph • top- $K_o = 3$ by predicted risk



9 LiDAR sectors, fan radius ~ 3.5 m

risk score r_{obs}

$s_1 : +40^\circ$	
$s_2 : -20^\circ$	
$s_3 : 0^\circ$	
$s_4 : -40^\circ$	
$s_5 : +60^\circ$	

c Risk-biased single-head attention



Repeated for 4 heads (128-D each), then aggregated to single-head output
 $\beta = 2.5$ scales the risk-aware attention bias